Docker

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Operating System Basics

1-What is an operating system?
software that manages the computer and helps programs run

2-Main parts of an operating system:
kernel, file system, memory management, process management.

3-What is the kernel?
The kernel is the core part of the operating system. It talks directly to the hardware.

4-Processes and running programs
A process is a program that is running right now.

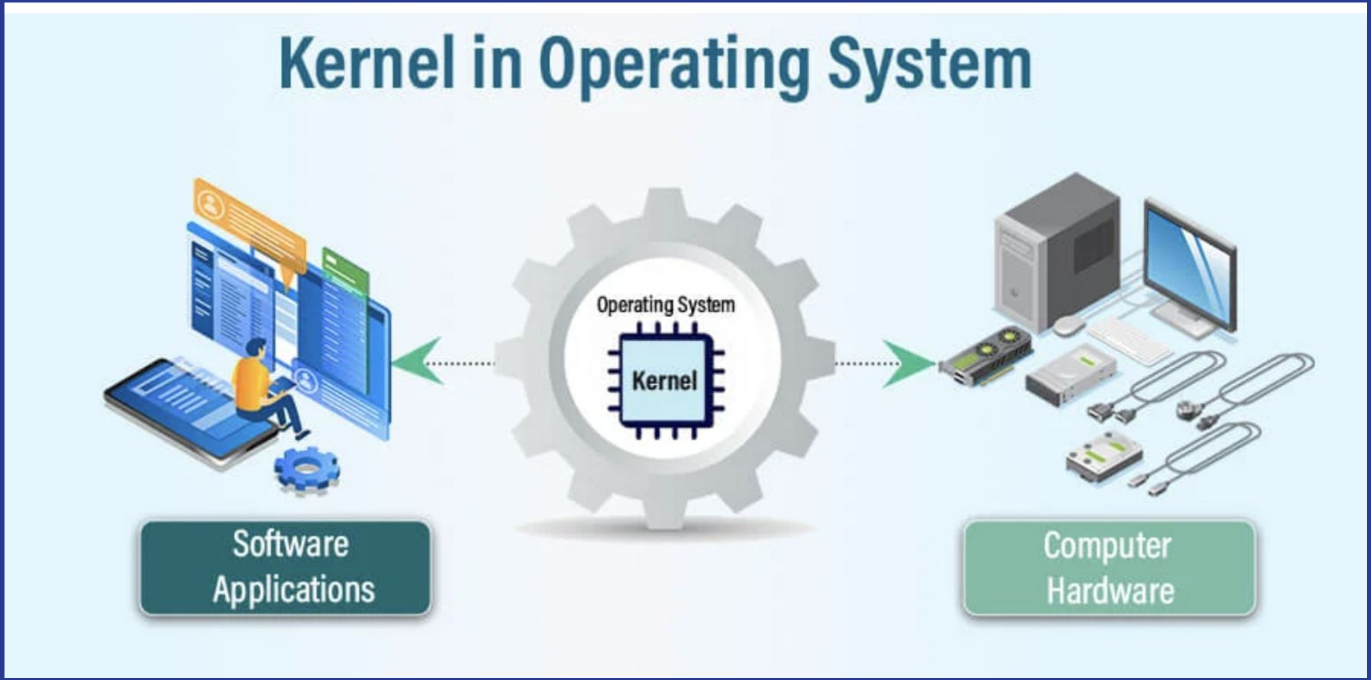
5-Filesystem basics
The file system is how the operating system stores and organizes files and folders.

6-Users and permissions
The operating system controls who can read, write, or run files.

Operating System Basics

Why Docker depends on the host OS?

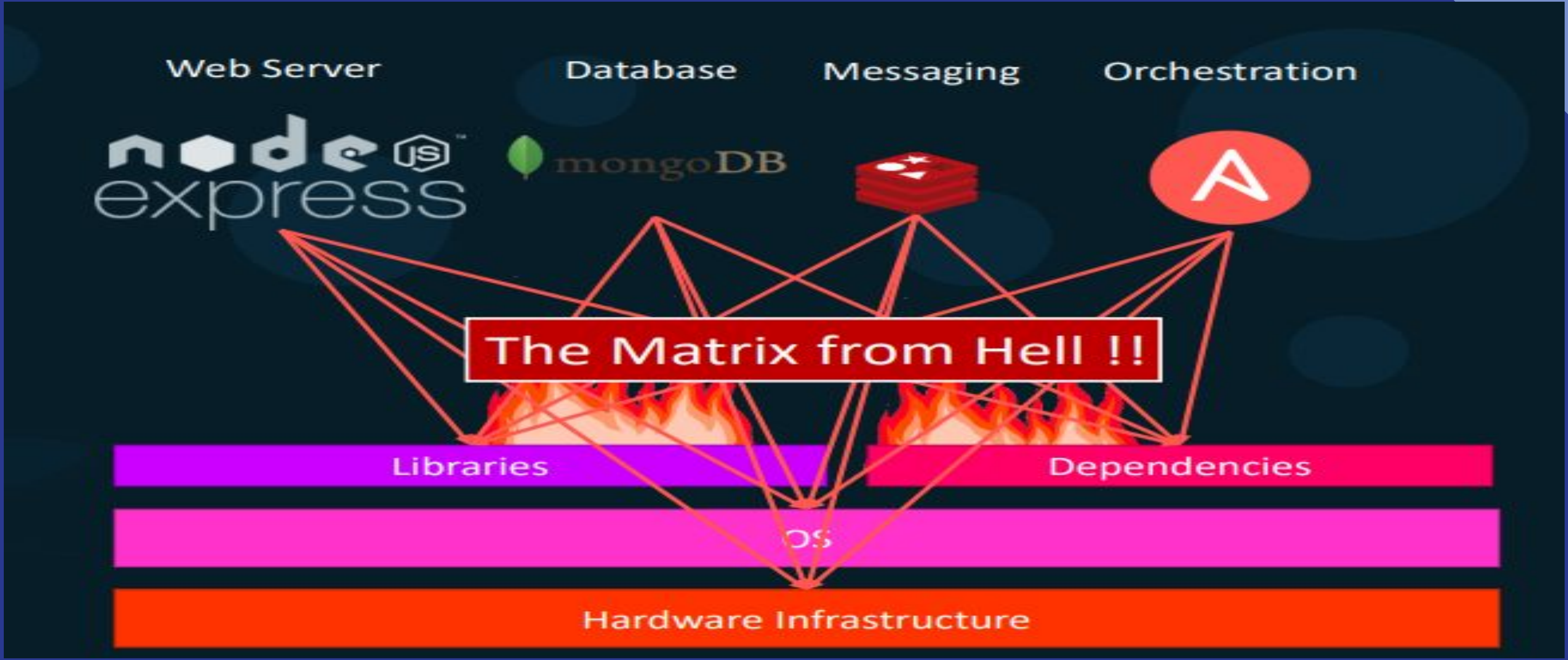
Docker containers use the host operating system kernel



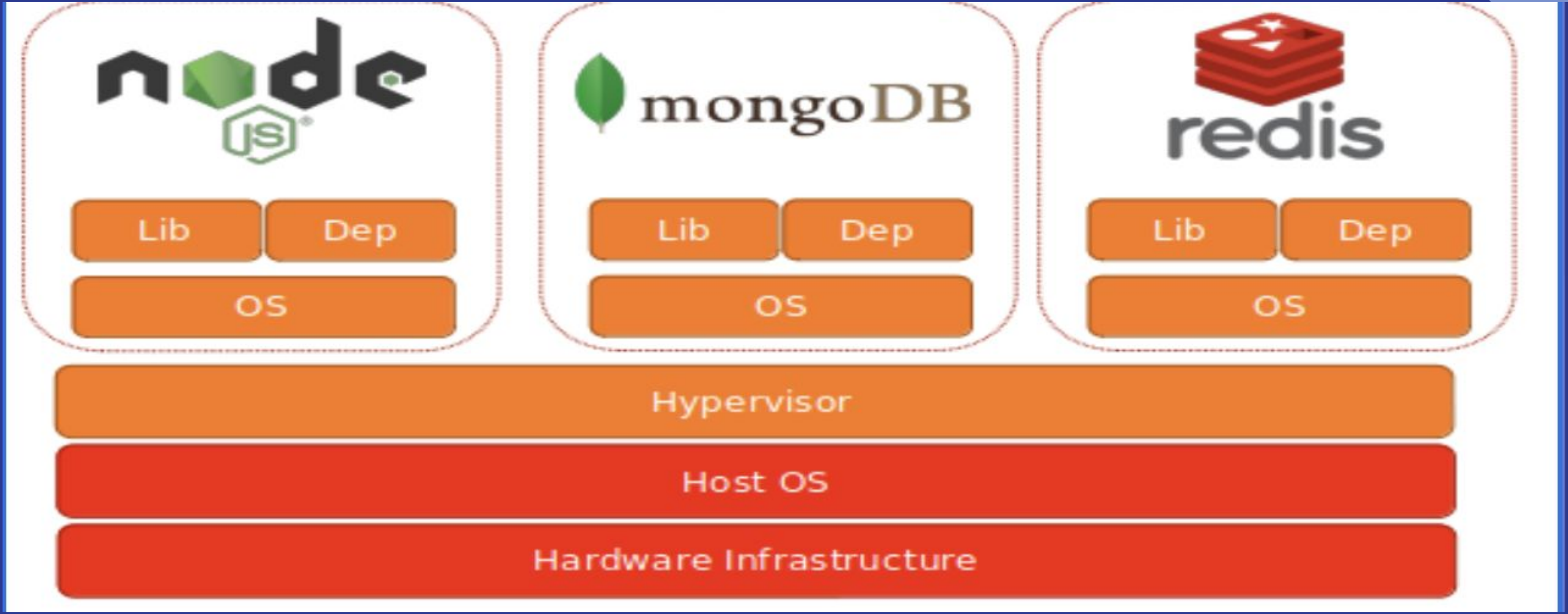
The Problem Before Containers

1-Different environments on each machine
2-Dependency version conflicts
3-Application works on one machine but fails on another
4-Hard setup for new developers
5-Manual installation steps
6-Problems moving app from dev to test to production
7-Resource waste with virtual machines
8-Difficult scaling and deployment
9-Hard rollback when updates fail
10-Dependency matrix from hell

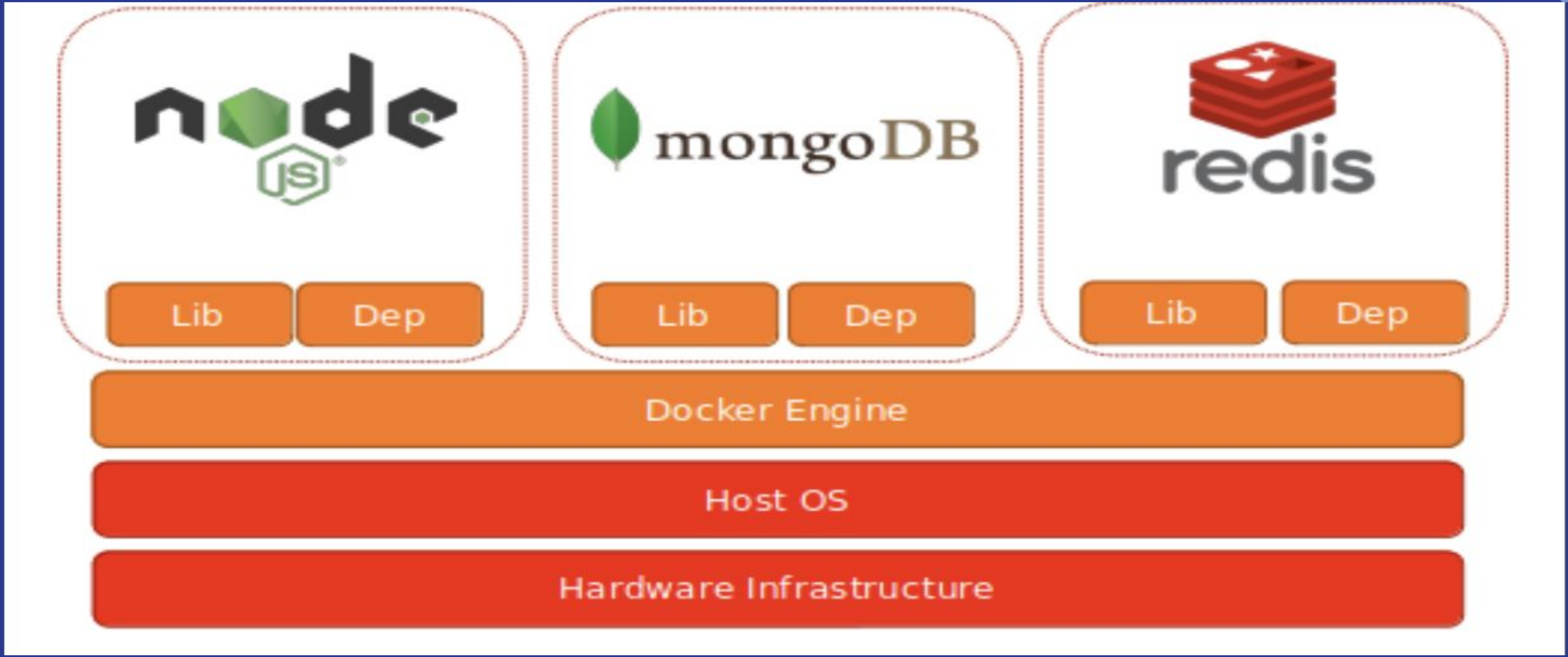
The Problem Before Containers



Virtualization Fundamentals

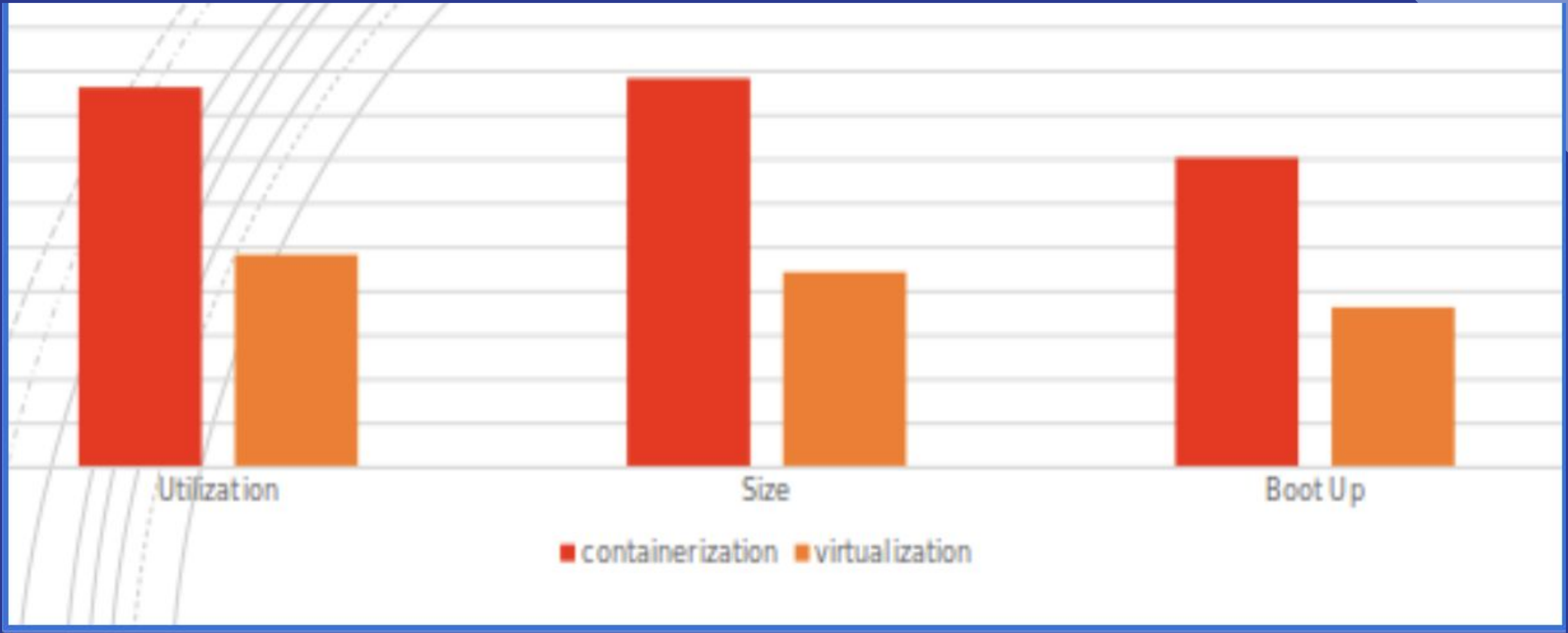


Containerization Fundamentals



Virtualization vs Containerization

Virtualization vs Containerization



5-Docker Overview

Docker is a platform around creating and running containers.

Docker is a tool designed to make it easier to create, deploy, and run applications by using containers

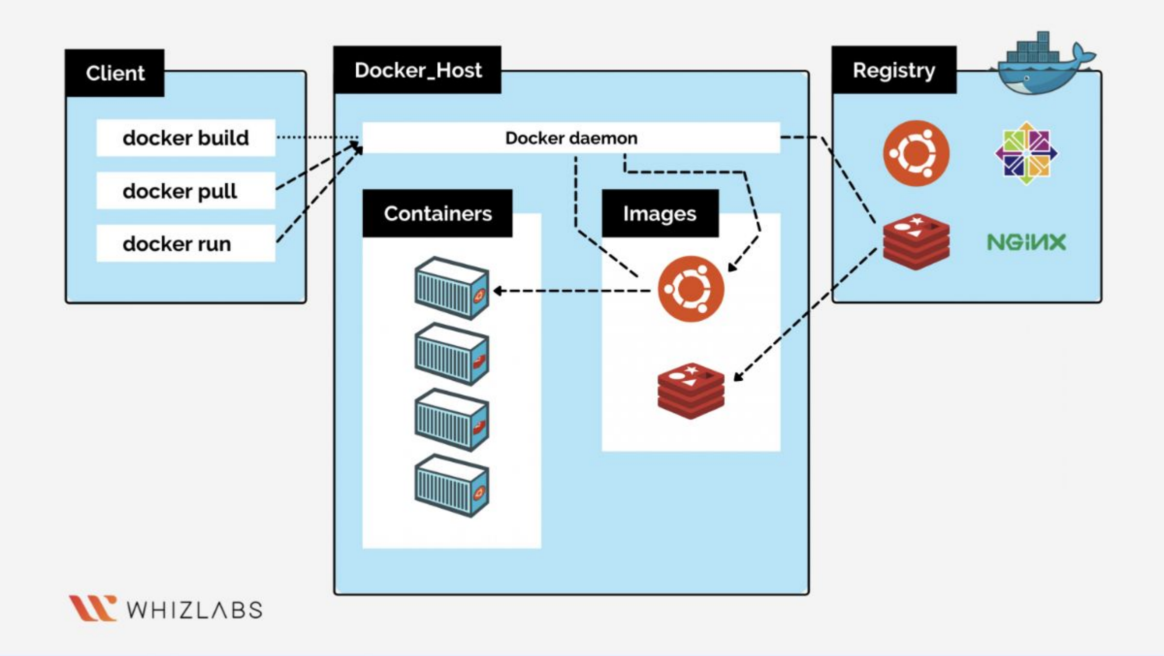
Docker Architecture

Docker client

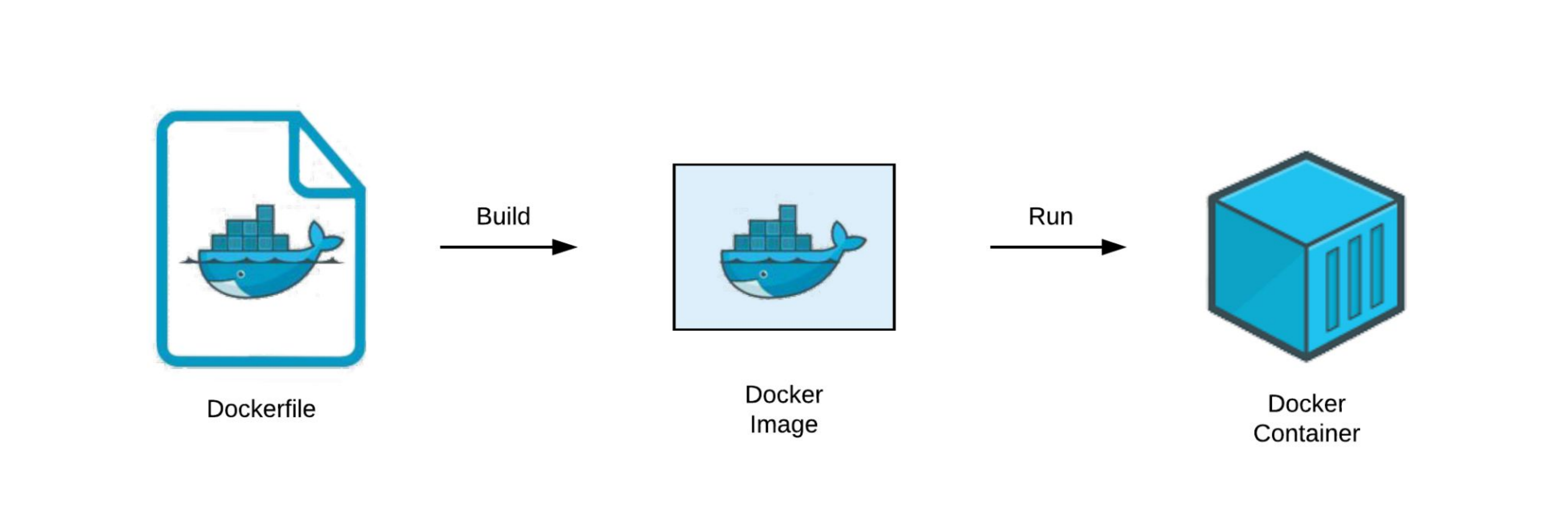
Docker Host

Docker daemon

Docker Registry



Core Docker Concepts



Installing Docker

https://docs.docker.com/engine/install/

Docker Images vs Docker Container

Essential Docker Commands

Run Container : docker run image-name or image-id

Overriding default command: docker run image-name or image-id <command>

Attached and Detached : docker run -d image-name or image-id

Essential Docker Commands

Port mapping : docker run -p localhost-port: container-port image-name or image-id

Stop all running container: docker stop $(docker ps)

Inspect container : docker inspect container-id

Stop container: docker stop container-id

Start container: docker start container-id

Start shell: Docker exec -it container-id

Essential Docker Commands

Docker exec container-id sh -c command

List all images: docker images

List all containers : docker ps -a

List all running containers : docker ps

Delete image: docker rmi image-name or image-id

Delete container : docker rm container-id

Problem 1

Run the container hello-world

Check the container status

Start the stopped container

Remove the container

Remove the image

Problem 2

Run container centos or ubuntu in an interactive mode

Run the following command in the container “echo docker”

Open a bash shell in the container and touch a file named hello-docker

Remove all stopped containers

Problem 3

Run the image nginx

Add html static files to the container and make sure they are accessible

Create html page in your machine and copy it to nginx start page

Lab 1

https://learn.kodekloud.com/user/courses/crash-course-docker-for-absolute-beginner/module/f4c089c7-24c8-488f-bb50-bbf844810374/lesson/904d093f-5547-4845-bb22-994e9e3f6c7a

Lab 2

https://learn.kodekloud.com/user/courses/crash-course-docker-for-absolute-beginner/module/f4c089c7-24c8-488f-bb50-bbf844810374/lesson/281f519a-dc55-4b0c-bd40-3e18f8b6c1f5

Lab 3

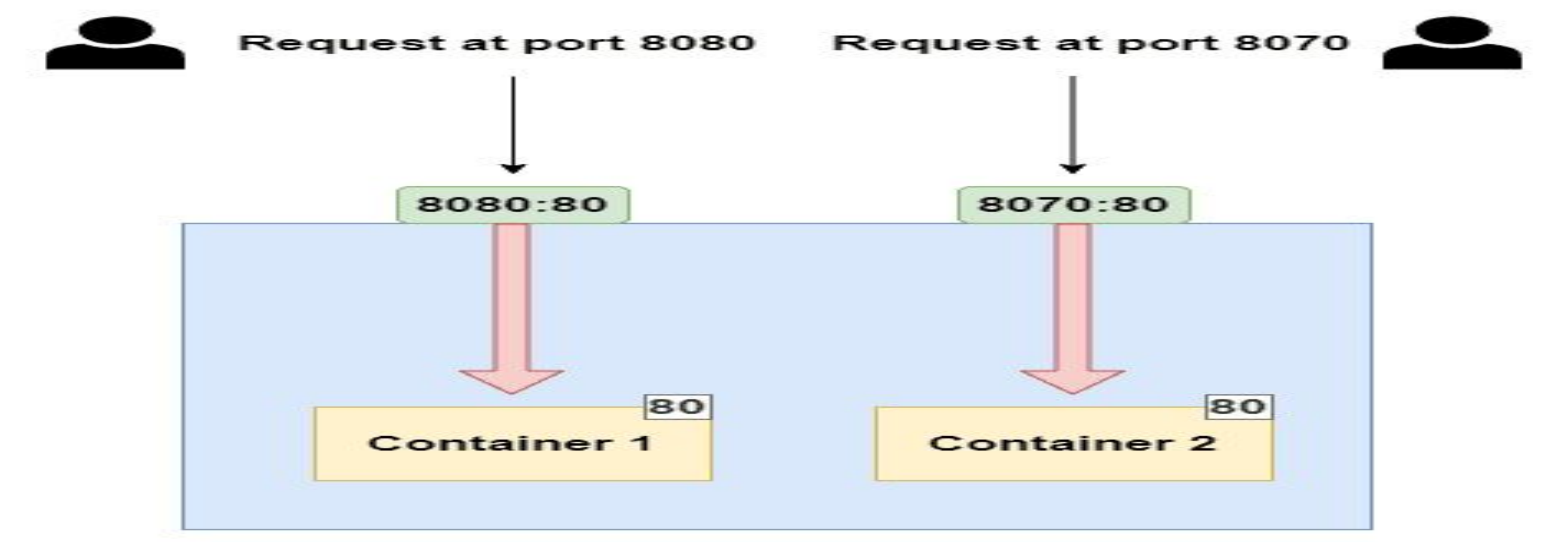
https://learn.kodekloud.com/user/courses/crash-course-docker-for-absolute-beginner/module/f4c089c7-24c8-488f-bb50-bbf844810374/lesson/3a8a2b0c-9c42-424d-9599-c44229fa0fab

Lab 4

https://learn.kodekloud.com/user/courses/crash-course-docker-for-absolute-beginner/module/f4c089c7-24c8-488f-bb50-bbf844810374/lesson/c2b63a97-2847-42f6-bad8-8004c61b36a0

Port Mapping

Docker run -p localport:containerport ima-name



Environment Variables

Run container of mysql image

Environment Variables

Docker run -e variable=value image-name

You can list all env :

docker exec -it container-id env
Docker inspect container-id

Lab1

https://learn.kodekloud.com/user/courses/docker-training-course-for-the-absolute-beginner/module/26faab43-a0ea-4355-9a94-f0bac957b507/lesson/6eab379d-70cc-4ae7-871e-b861d5be6883

Docker Networking

Docker Networking

List all networks: docker ls network

Create network :

docker network create \
--driver your driver \
--subnet 182.18.0.0/16 \
> network-name

Run container in specific network :

Docker run –network=networkname image-name

Problem 4

Create network custom-network with bridge driver and subnet 182.18.0.0/16

List all network

Run httpd container in custom-network

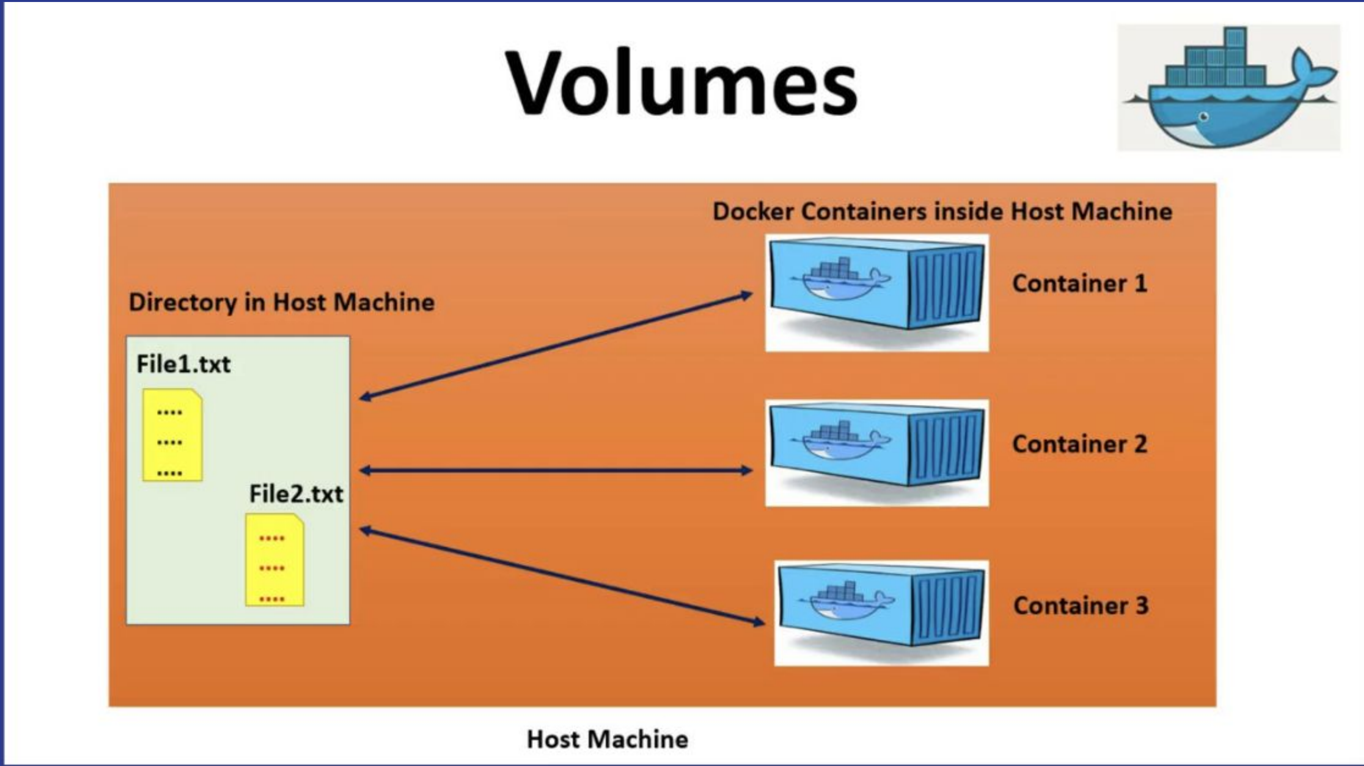
Get the ip of httpd container

Get the network of container

Lab 3

https://learn.kodekloud.com/user/courses/docker-training-course-for-the-absolute-beginner/module/7dedf6e6-6dca-4f9a-b120-0807d6cc9194/lesson/b7c5daf5-bf08-4b43-8c68-9c959be586f6

Docker Volumes and Storage



Docker Volumes and Storage

```
docker volume create
```

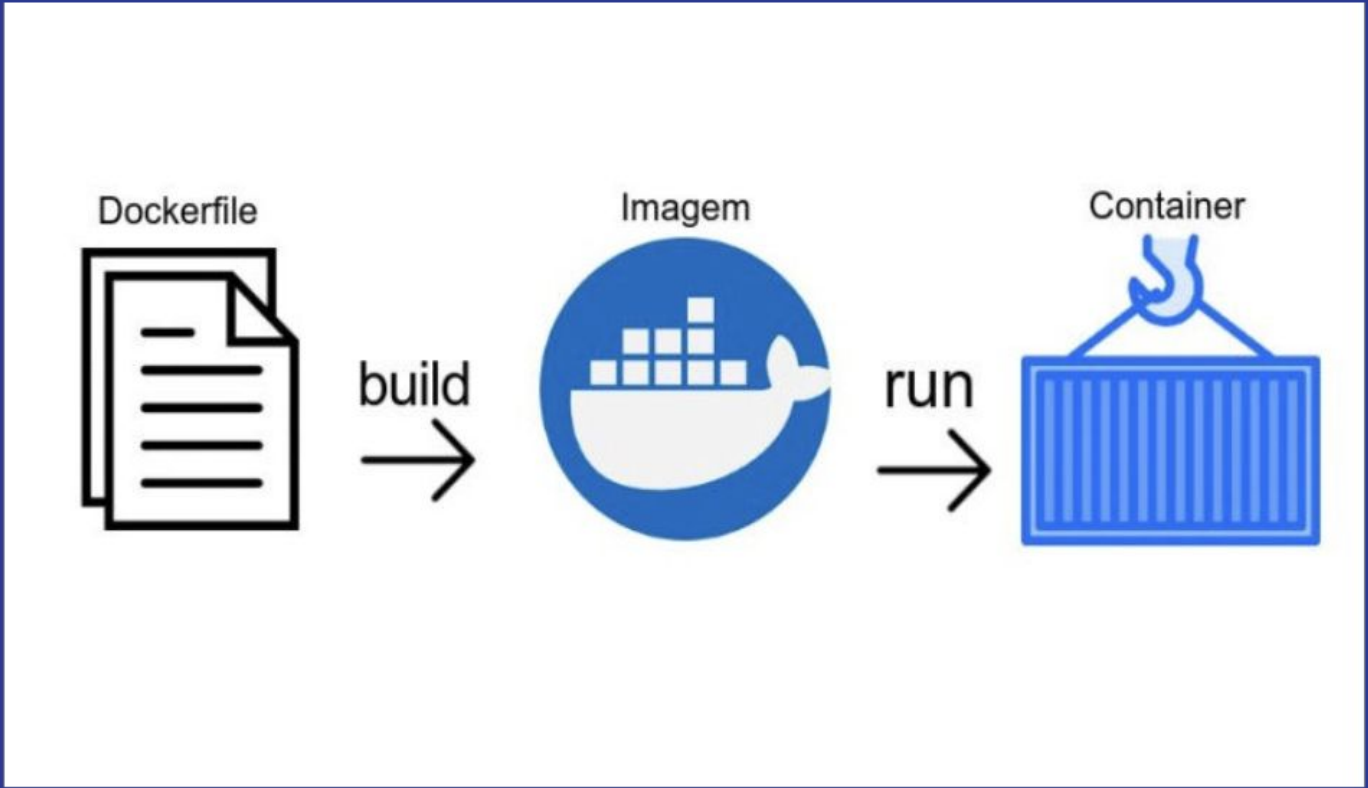
Docker ls volume

docker run -v localpath:/pathincontainer image-name

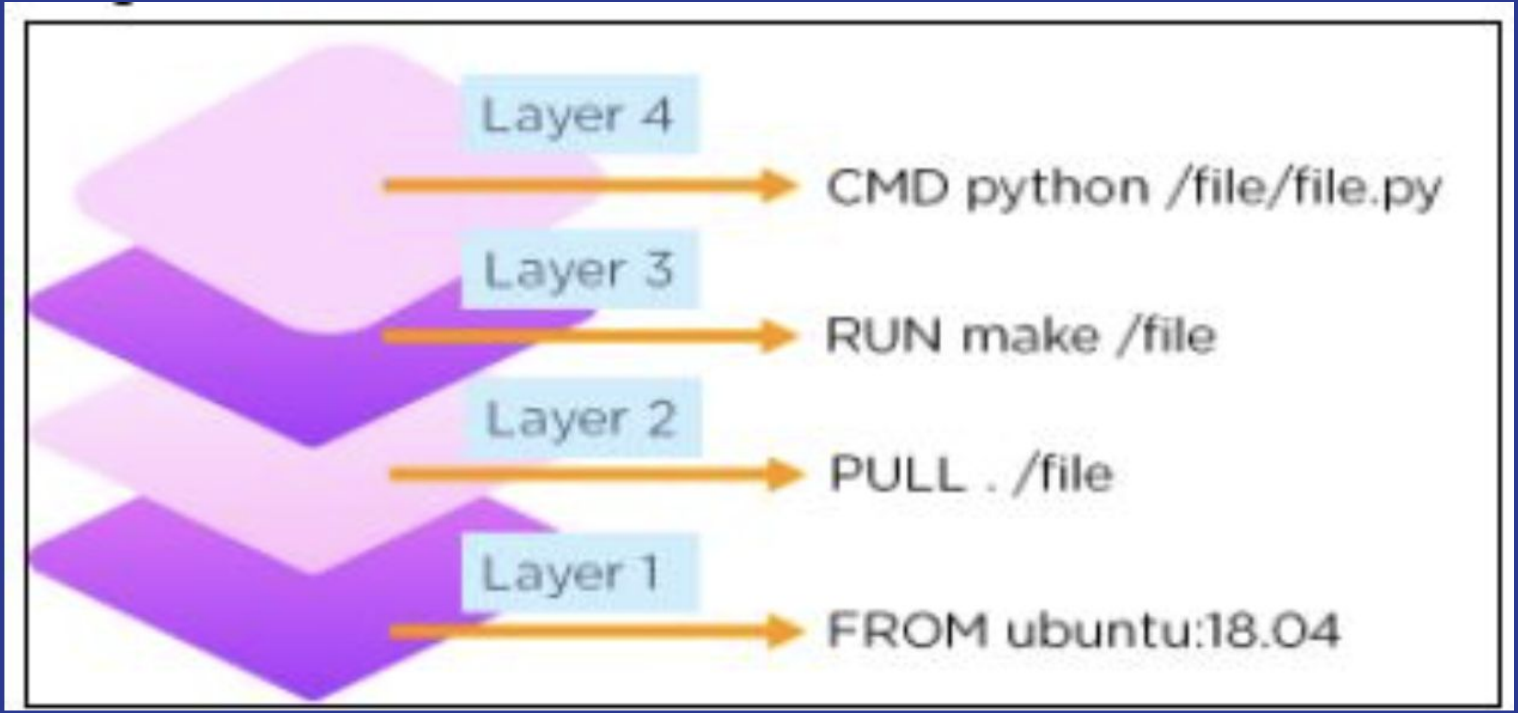
Working with Databases in Containers

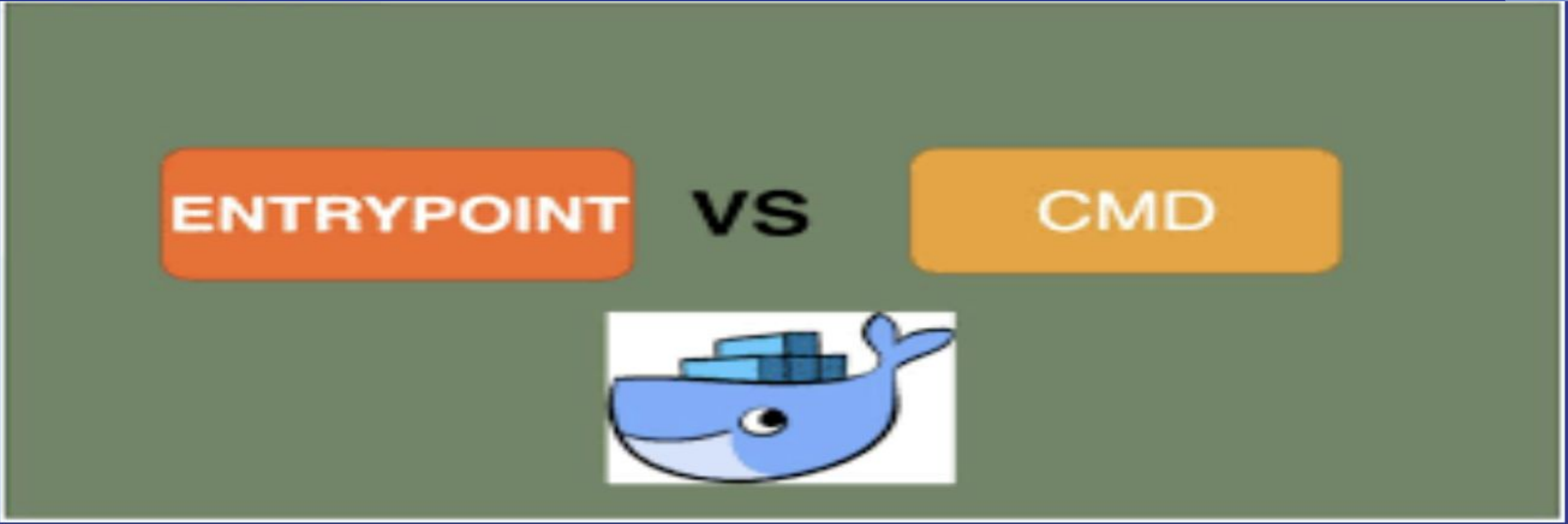
Create postgres container and connect from pd_admin container

Dockerfile Basics



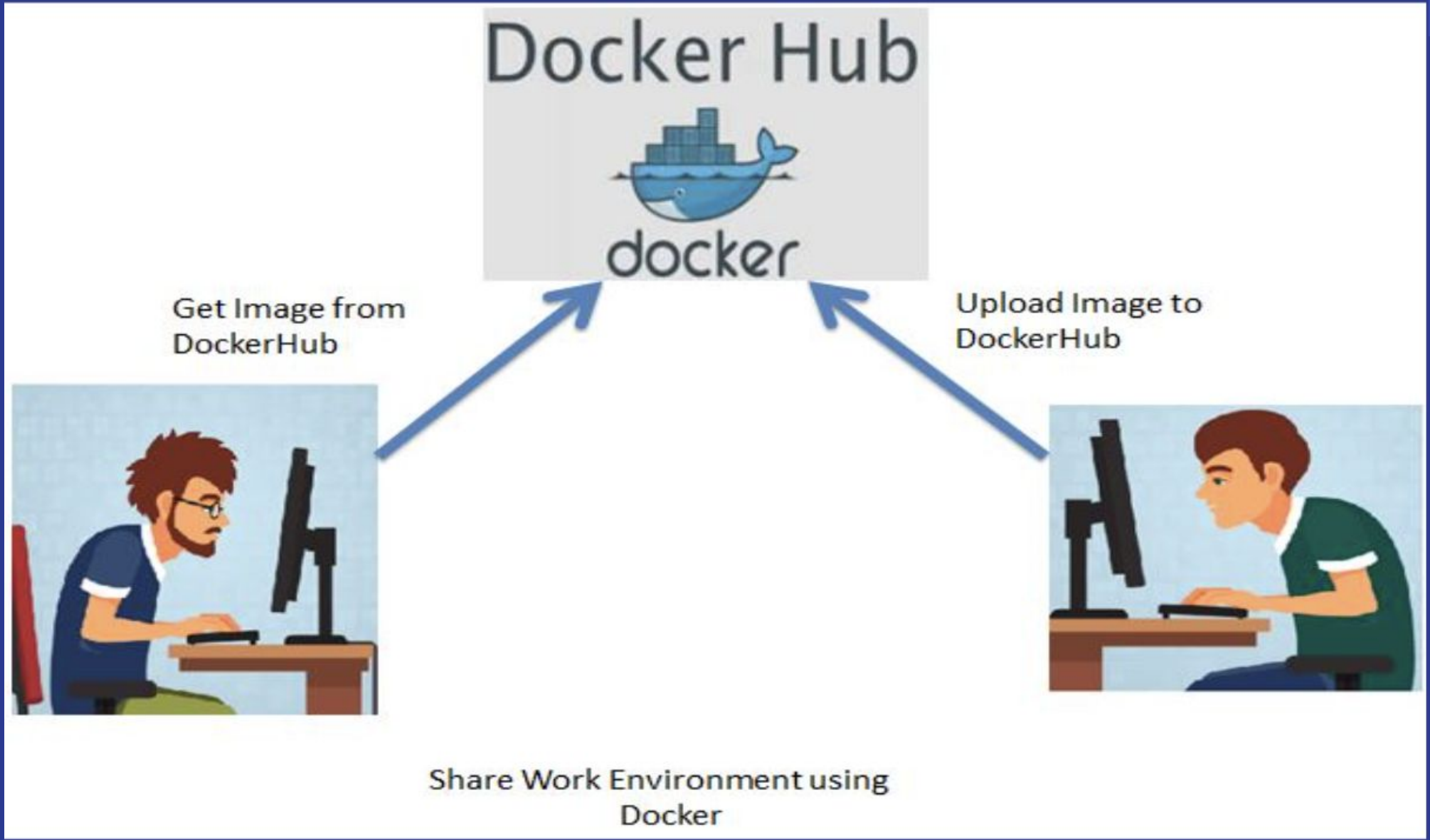
Dockerfile Basics




ENTRYPOINT
VS
CMD

Docker Compose

Docker Registries



Publishing Images

Docker login

Docker images

Docker tag img-name

dockerhub_username/reponame:tag

Image Optimization

Smaller

Faster to build

More secure